

DUDINKA, Russian Federation

WMO: 230740

Lat: 69.400N

Lon: 86.167E

Elev: 62

StdP: 14.66

Time Zone: 7.00 (E07)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-50.2	-46.8	-55.1	0.2	-50.3	-51.5	0.3	-46.8	31.5	5.8	27.6	7.5	3.9	110	0.947

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	77.0	63.4	71.8	61.0	67.4	58.5	64.9	73.8	62.1	70.0	59.5	66.2	9.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	81.0	68.7	58.5	73.3	66.2	56.0	66.9	63.7	30.0	73.7	28.0	69.8	26.2	66.8	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	24.3	21.4	DB	-53.4	81.7	5.9	3.7	-57.6	84.4	-61.0	86.5	-64.3	88.5	-68.5	91.2	(4)
(5)			WB	-53.4	66.4	5.9	2.8	-57.6	68.4	-61.0	70.0	-64.3	71.6	-68.5	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	-15.0	-16.7	-4.9	7.7	24.1	46.5	56.9	51.9	41.1	19.3	-4.6	-12.0		
	DBStd	29.84	17.57	17.83	17.84	15.25	10.80	10.23	7.41	6.82	7.45	14.06	17.29	17.88		
	HDD50	12714	2014	1867	1701	1269	807	189	15	57	281	952	1639	1923		
	HDD65	17776	2479	2287	2166	1719	1269	562	273	409	718	1417	2089	2388		
	CDD50	446	0	0	0	0	2	85	229	117	13	0	0	0		
	CDD65	33	0	0	0	0	0	7	22	4	0	0	0	0		
	CDH74	241	0	0	0	0	1	59	161	20	0	0	0	0		
	CDH80	33	0	0	0	0	0	9	24	0	0	0	0	0		
	WSAvg	10.6	10.6	9.5	10.3	10.6	11.1	10.7	11.1	10.3	10.7	10.9	9.7	11.1		
	PrecAvg	20.1	1.4	1.3	1.3	1.3	1.3	1.3	2.2	2.3	1.6	2.2	1.7	1.6		
	PrecMax	25.4	3.2	3.0	2.8	2.7	2.5	3.2	10.3	3.8	3.9	3.8	3.2	2.9		
(15) Precipitation	PrecMin	14.7	0.2	0.3	0.3	0.1	0.1	0.1	0.0	0.7	0.6	0.2	0.2	0.0		
	PrecStd	3.1	0.7	0.7	0.8	0.7	0.6	0.9	2.1	0.9	0.8	0.7	0.7	0.7		
	DB	25.8	24.6	33.4	41.8	60.5	82.0	83.1	77.1	66.9	50.1	34.0	26.9			
	MCWB	25.0	24.0	30.7	36.3	50.3	63.0	65.6	65.8	56.7	47.1	32.6	26.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	17.4	19.2	28.1	37.3	43.9	74.4	79.0	70.3	58.4	41.0	30.4	23.3			
	2% MCWB	16.3	18.3	26.7	33.8	38.7	59.7	64.3	61.6	52.2	39.6	29.4	22.5			
	5% DB	12.4	13.8	24.0	34.4	40.2	68.5	74.1	66.2	54.7	38.7	24.5	17.9			
	5% MCWB	11.2	12.7	22.8	32.6	36.2	57.3	62.7	59.2	50.1	37.5	23.4	16.9			
	10% DB	7.8	9.0	19.7	30.8	37.0	62.9	69.3	62.6	51.5	35.2	19.5	12.7			
	10% MCWB	6.8	8.1	18.6	28.8	34.6	53.7	59.6	56.9	47.7	33.9	18.4	11.6			
	0.4% WB	25.2	23.8	31.5	36.8	50.4	65.3	68.4	67.3	59.4	47.0	33.3	26.2			
	0.4% MCDB	26.0	24.3	33.1	41.1	60.1	76.8	77.5	74.8	66.6	49.5	34.1	26.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	16.5	18.3	26.9	34.5	39.3	61.5	66.0	62.8	53.7	39.8	29.4	22.5			
	2% MCDB	17.3	19.3	28.2	36.9	43.1	72.7	75.3	68.4	57.1	40.7	30.6	23.3			
	5% WB	11.4	12.9	22.7	32.7	36.8	58.0	63.6	59.9	50.3	37.6	23.6	16.9			
	5% MCDB	12.3	13.9	23.7	34.5	39.5	68.2	72.4	65.0	54.1	38.8	24.6	17.8			
	10% WB	6.8	8.0	18.6	28.9	34.7	53.9	61.0	57.4	47.8	34.2	18.5	11.7			
	10% MCDB	7.7	9.0	19.7	30.7	36.8	62.7	68.3	62.0	50.9	35.1	19.6	12.7			
	MDBR	12.6	13.1	16.0	16.2	11.0	12.1	13.1	11.7	8.9	9.3	12.8	12.1			
	5% DB	15.1	17.4	14.6	11.0	11.6	21.7	19.2	17.9	14.4	6.3	14.7	15.4			
(36) Mean Daily Temperature Range	MCWBR	14.4	17.0	14.2	9.6	8.2	10.9	8.9	10.0	9.1	5.8	14.5	15.2			
	MCDBR	14.6	17.6	14.7	11.1	11.1	19.6	17.0	15.2	12.6	6.4	14.8	15.6			
	MCWBR	14.6	17.4	14.3	10.1	8.5	11.2	9.0	9.4	8.9	5.9	14.7	15.4			
	MCDBR	14.6	17.4	14.3	10.1	8.5	11.2	9.0	9.4	8.9	5.9	14.7	15.4			
(40) Clear-Sky Solar Irradiance	taub	0.216	0.249	0.305	0.328	0.353	0.370	0.400	0.393	0.324	0.250	0.201	N/A			
	taud	1.856	1.959	1.997	2.040	2.097	2.344	2.290	2.258	2.401	2.084	1.883	N/A			
	Ebn at Noon	5	147	214	251	259	261	246	230	220	147	4	0			
	Edh at Noon	2	15	29	37	40	33	34	31	20	12	1	0			
(44) All-Sky Solar Radiation	RadAvg	4	110	607	1300	1808	1698	1589	995	500	157	17	0			
	RadStd	0	15	62	106	135	146	221	115	58	28	2	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page